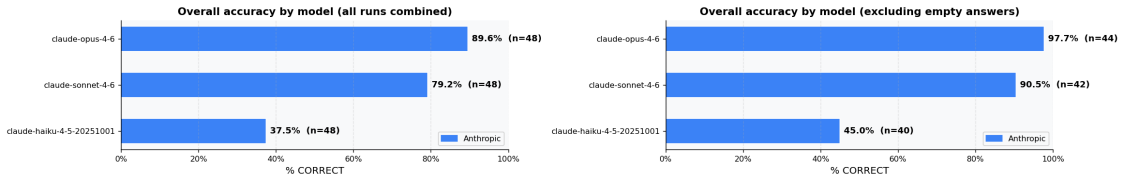
results_analysis_v2

March 14, 2026

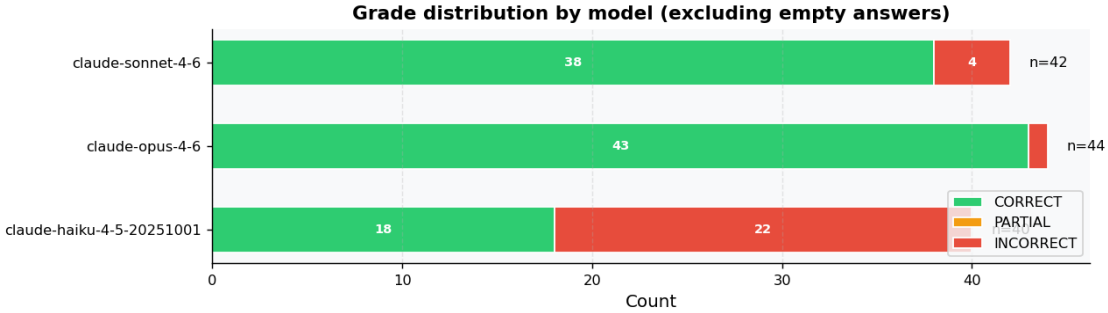
1 TinyBench — Evaluation Results Analysis

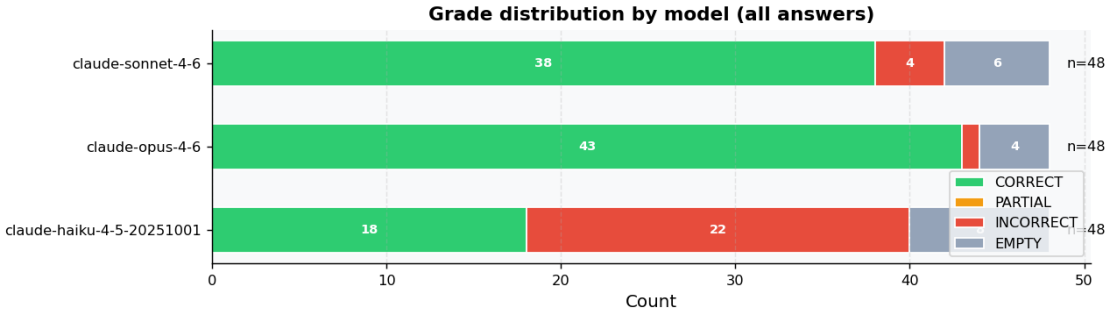
```
Loaded 2 result file(s):
  ../data/results/graded/eval_results_tools_get_semantic_context_run_python_grad
ed.json  →  run='semantic'
  ../data/results/run_03_13_2026/eval_python_only_graded.json  →
run='python_only'

144 total records  |  3 models  |  2 runs  |  2 templates
Empty answers: 18
Records after excluding empty answers: 126 / 144
Runs present: ['python_only', 'semantic']
```

1.1 1 — Overall Accuracy by Model

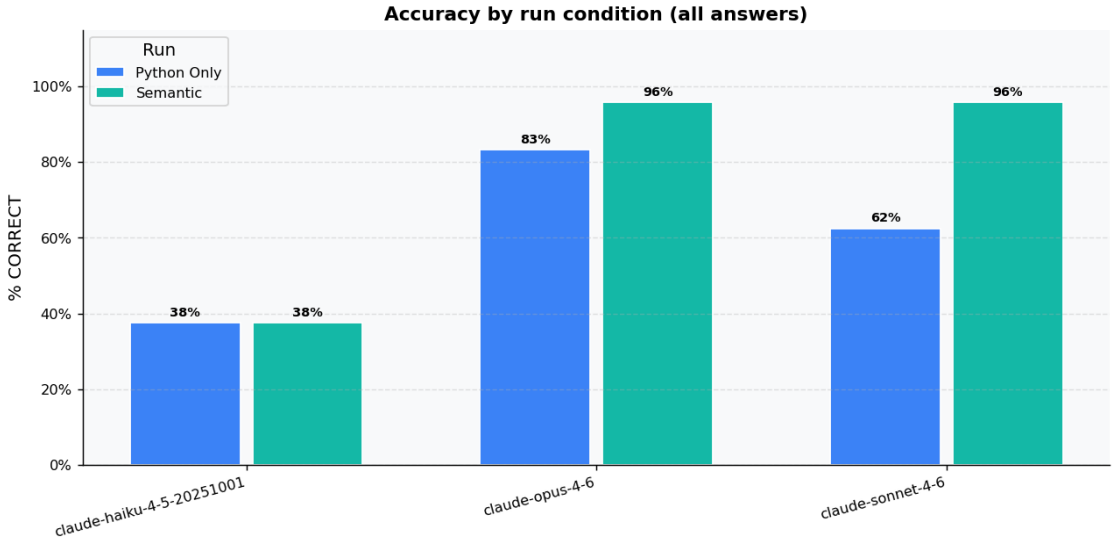


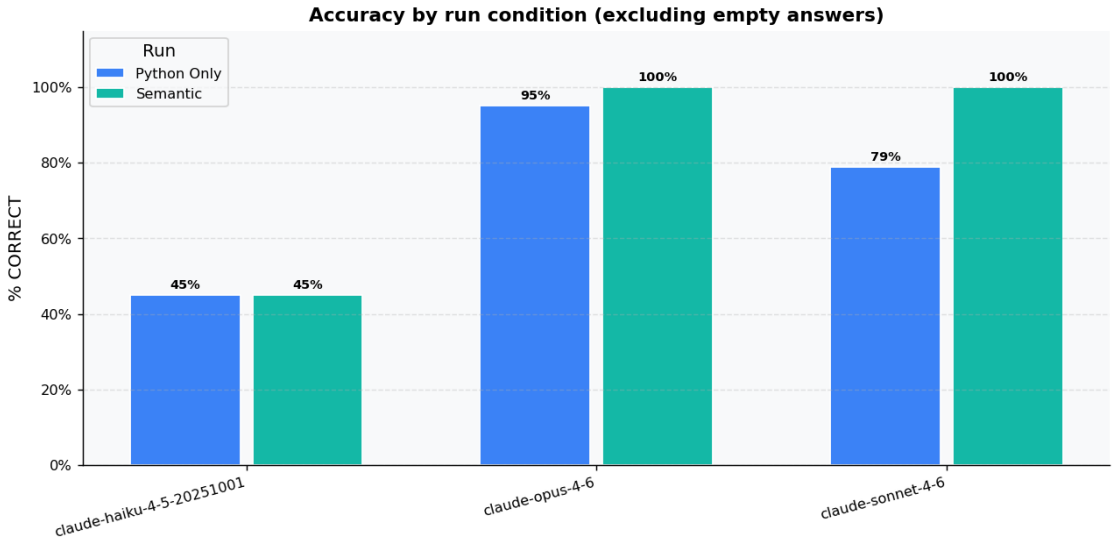
1.2 2 — Grade Distribution

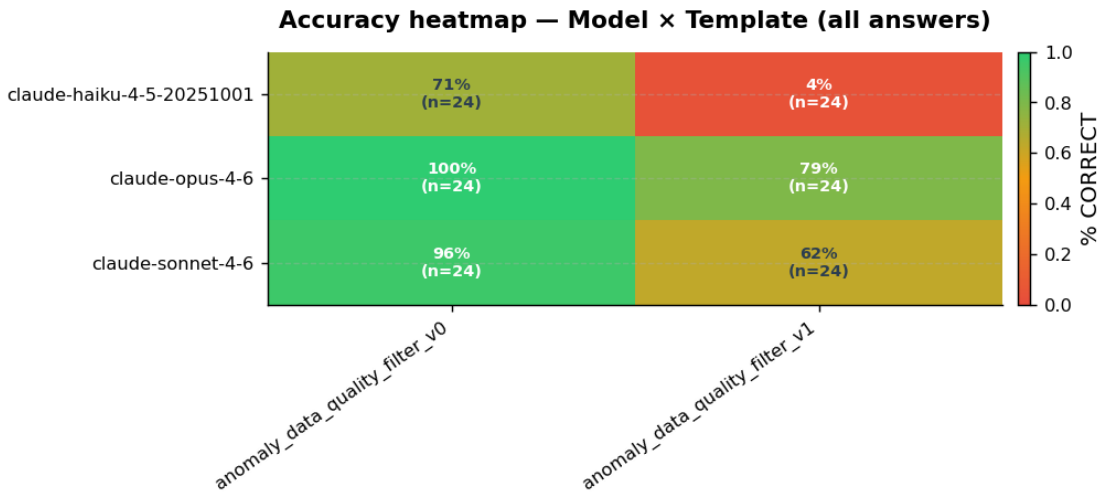


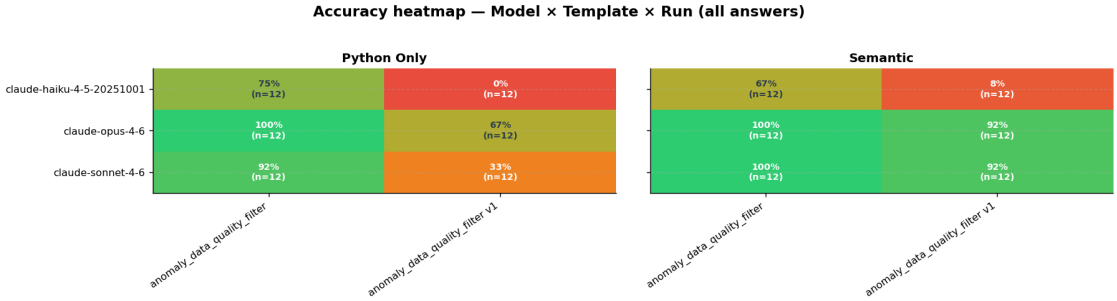


1.3 3 — Accuracy by Run Condition

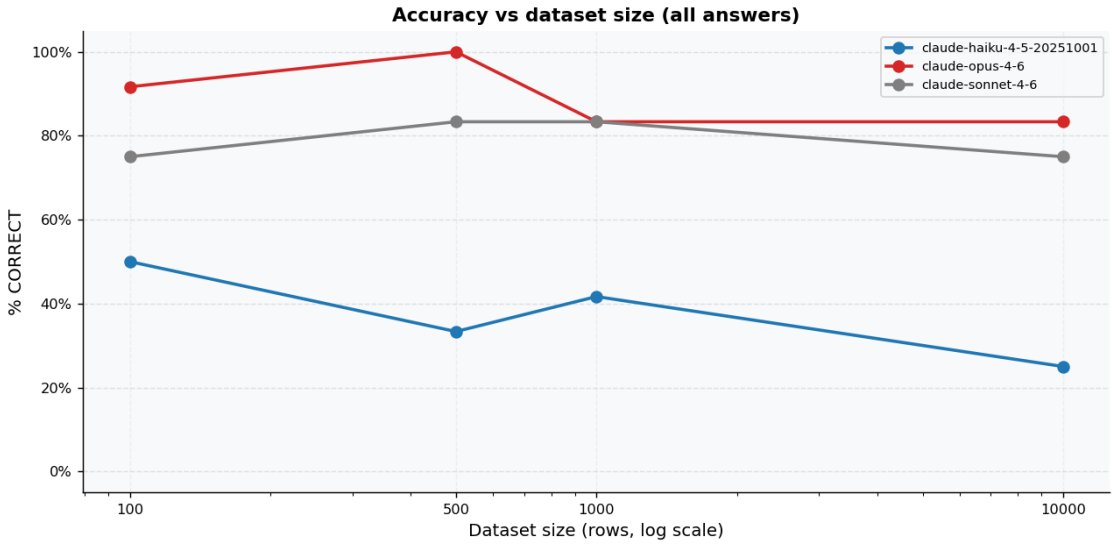


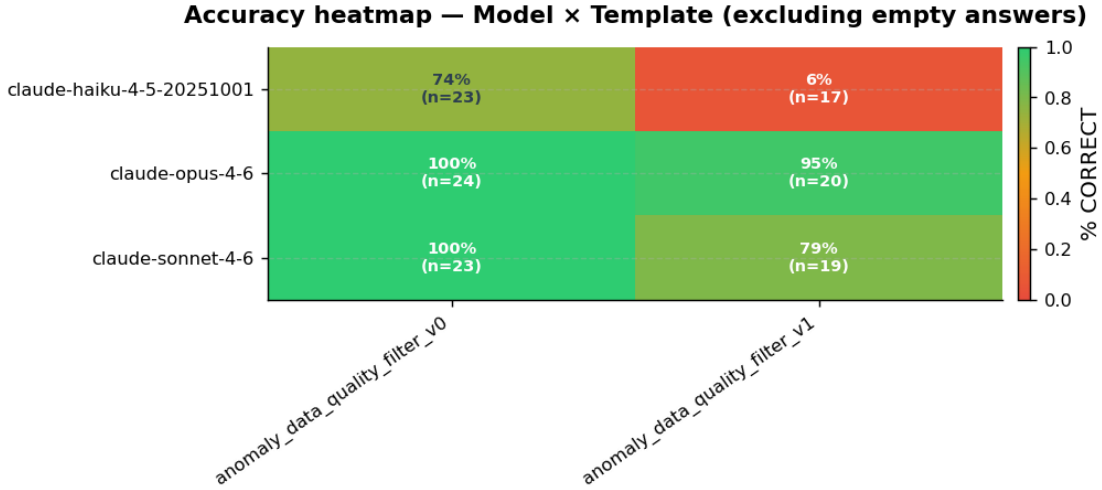

Accuracy by run condition (excluding empty answers)
Run
Python Only
Semantic
% CORRECT
100%
80%
60%
40%
20%
0%
45%
45%
95%
100%
79%
100%
claude-haiku-4-5-20251001
claude-opus-4-6
claude-sonnet-4-6

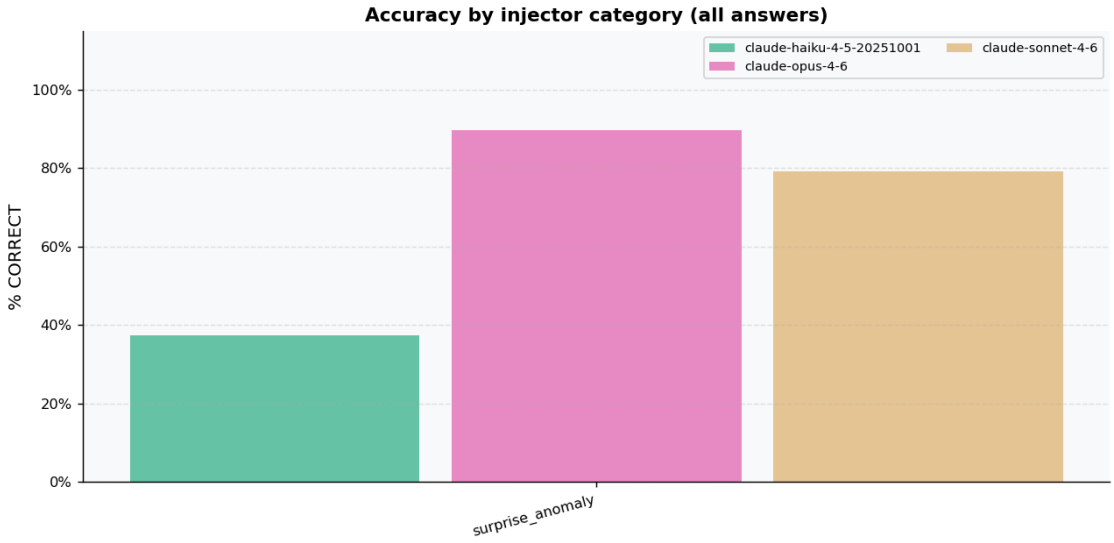

Accuracy heatmap — Model × Template (all answers)
claude-haiku-4-5-20251001
claude-opus-4-6
claude-sonnet-4-6
71% (n=24)
4% (n=24)
100% (n=24)
79% (n=24)
96% (n=24)
62% (n=24)
anomaly_data_quality_filter_v0
anomaly_data_quality_filter_v1
% CORRECT
1.0
0.8
0.6
0.4
0.2
0.0


Accuracy heatmap — Model × Template × Run (all answers)
Python Only
Semantic
claude-haiku-4-5-20251001
claude-opus-4-6
claude-sonnet-4-6
75% (n=12)
0% (n=12)
100% (n=12)
67% (n=12)
92% (n=12)
33% (n=12)
67% (n=12)
8% (n=12)
100% (n=12)
92% (n=12)
100% (n=12)
92% (n=12)
anomaly_data_quality_filter
anomaly_data_quality_filter v1
anomaly_data_quality_filter
anomaly_data_quality_filter v1

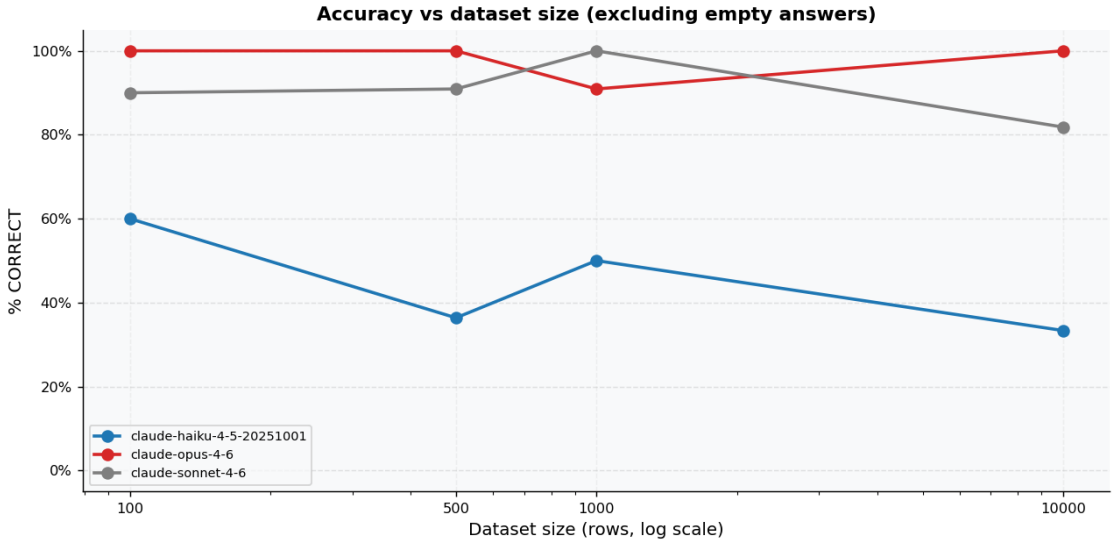
1.4 4 — Accuracy Heatmap: Model × Template



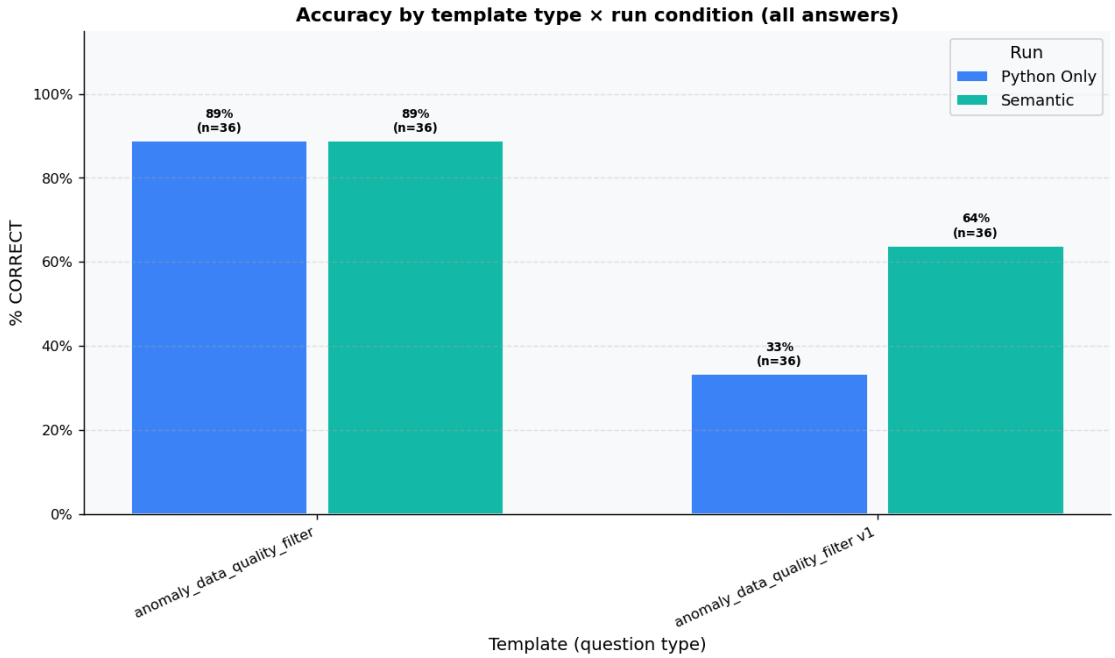


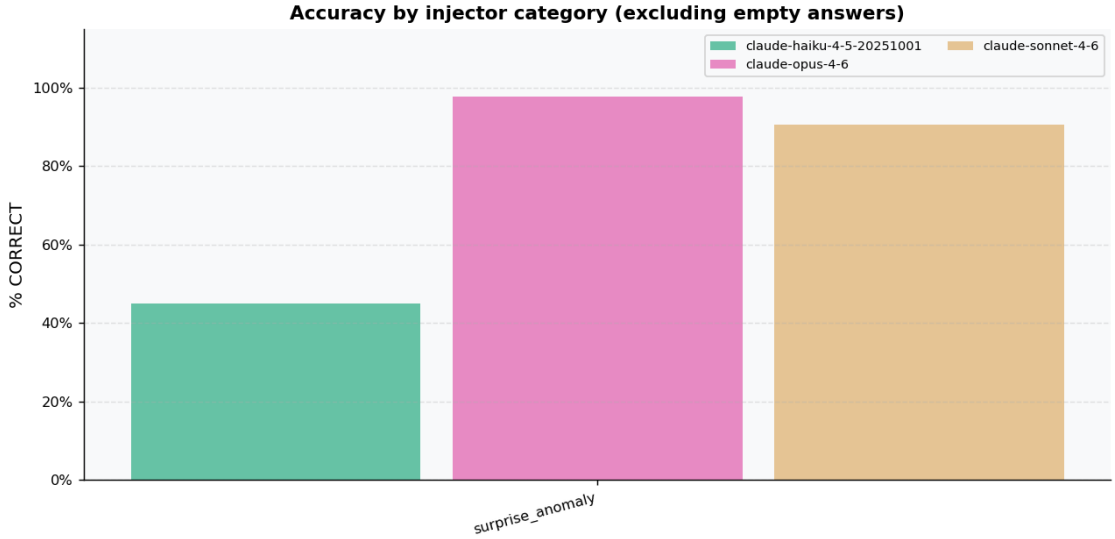


1.5 5 — Accuracy by Dataset Size



1.6 6 — Accuracy by Injector Category

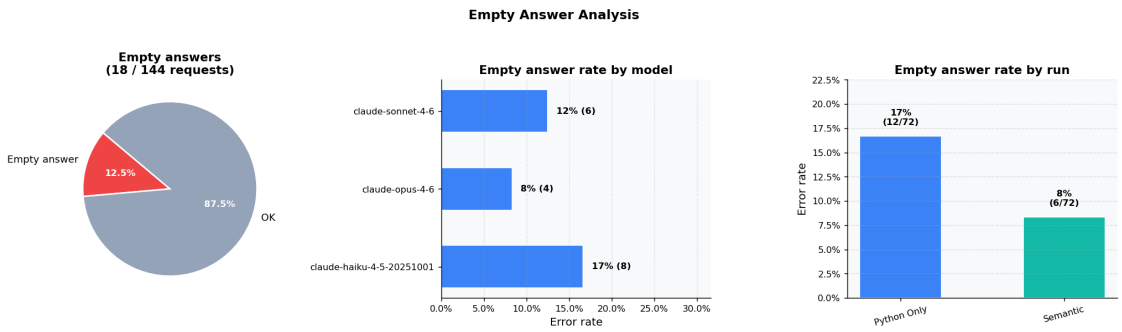




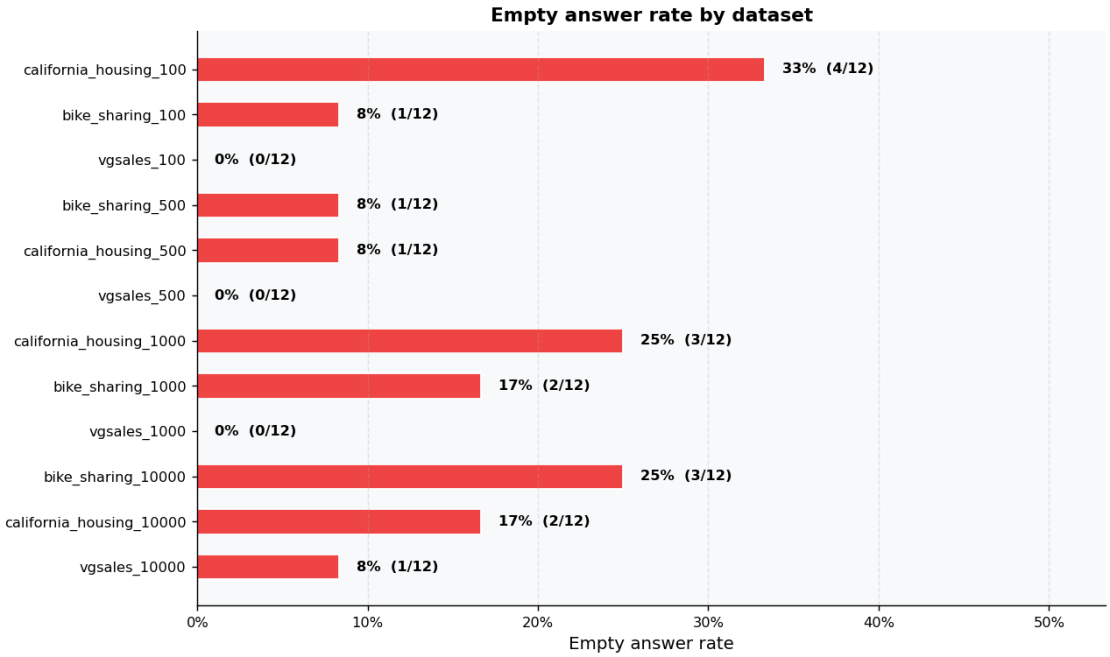
1.7 7 — Empty Answer Analysis

```
Empty answers: 18 / 144 (12.5%)
Affected datasets: ['bike_sharing_100', 'bike_sharing_1000',
```

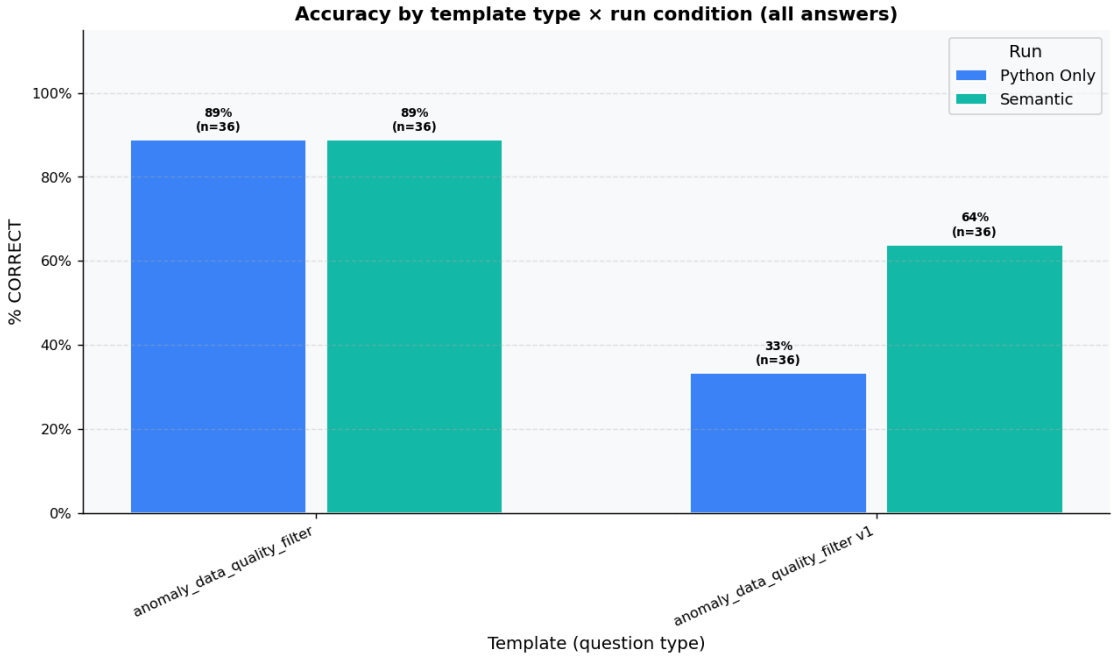
```
'bike_sharing_10000', 'bike_sharing_500', 'california_housing_100',
'california_housing_1000', 'california_housing_10000', 'california_housing_500',
'vgsales_10000']
```

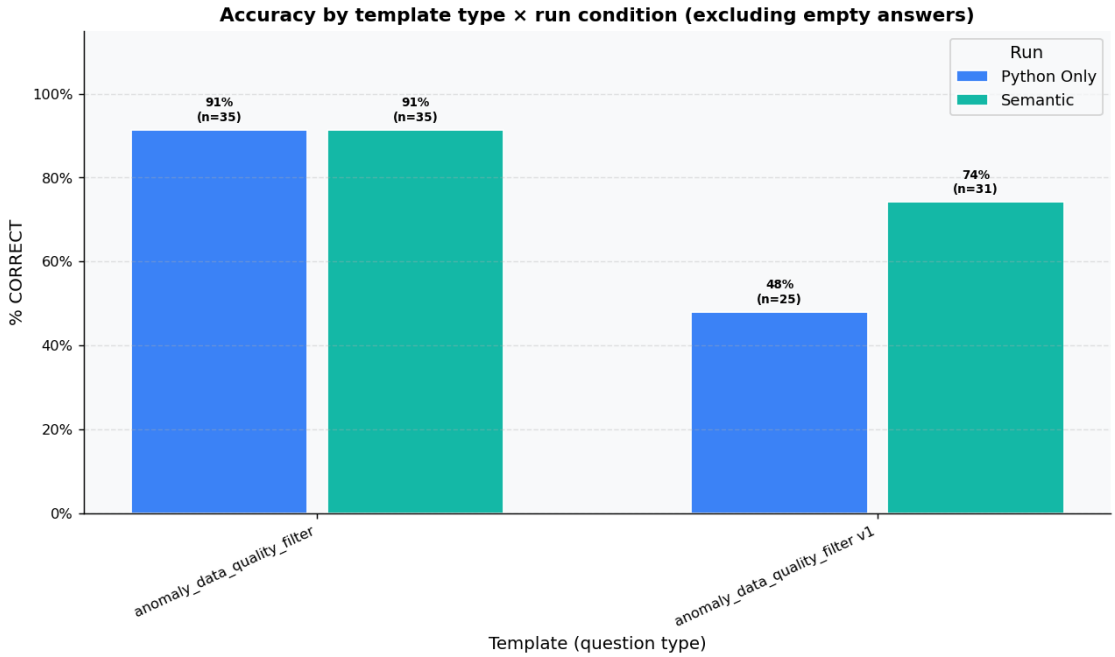



```
Empty answers by model × run:
                                      errors  total rate
model                     run
claude-haiku-4-5-20251001 python_only       4     24  17%
                          semantic          4     24  17%
claude-opus-4-6           python_only       3     24  12%
                          semantic          1     24   4%
claude-sonnet-4-6         python_only       5     24  21%
                          semantic          1     24   4%

Empty answers by dataset:

                         errors  total  rate rate_fmt
dataset
california_housing_100        4     12  0.33      33%
bike_sharing_100              1     12  0.08       8%
vgsales_100                   0     12  0.00       0%
bike_sharing_500              1     12  0.08       8%
california_housing_500        1     12  0.08       8%
vgsales_500                   0     12  0.00       0%
california_housing_1000       3     12  0.25      25%
bike_sharing_1000             2     12  0.17      17%
vgsales_1000                  0     12  0.00       0%
bike_sharing_10000            3     12  0.25      25%
california_housing_10000      2     12  0.17      17%
vgsales_10000                 1     12  0.08       8%

Datasets with errors: 9 / 12
```



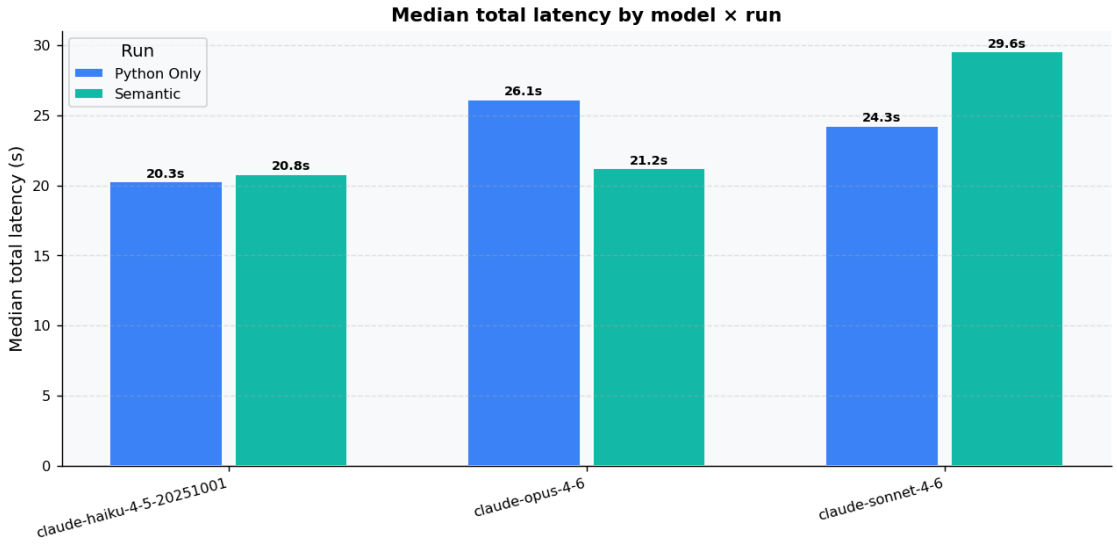
1.8 8 — Template × Run Accuracy



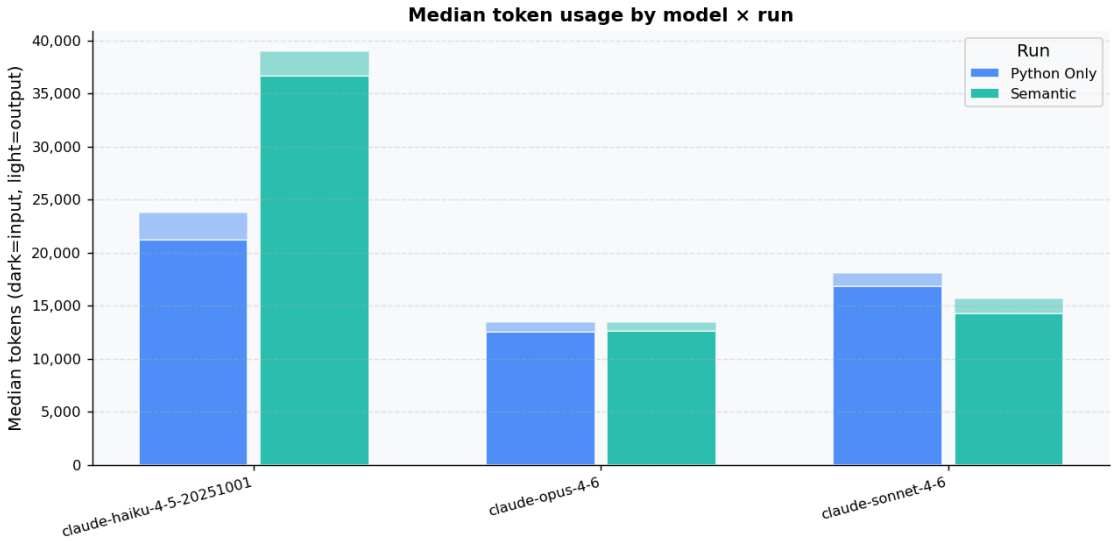


1.9 9 — Latency & Token Metrics

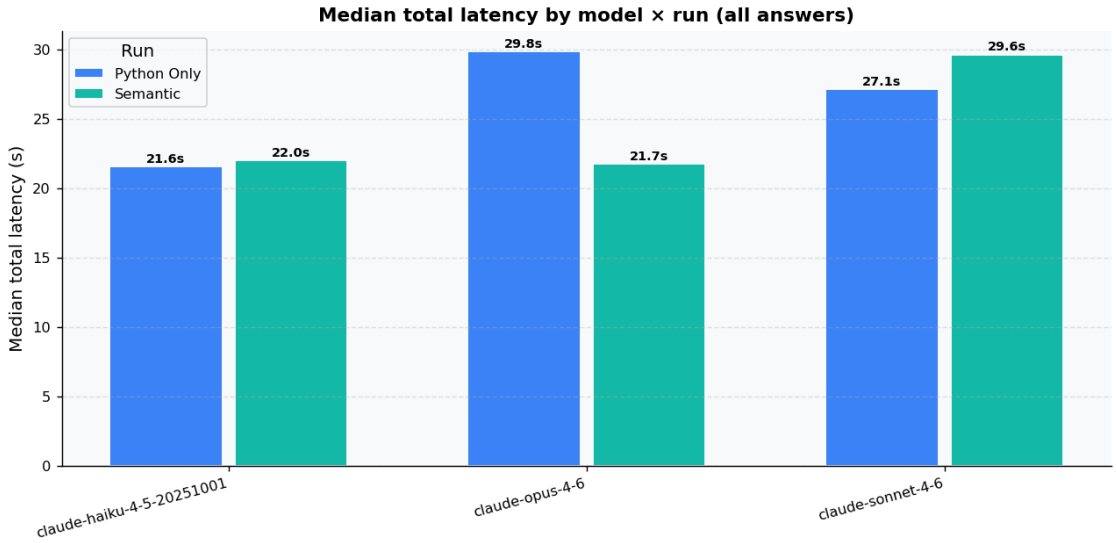
All charts use **median** values (robust to outliers). Error entries are excluded. Metrics: `total_latency_s` (wall-clock), `ttft_s` (time-to-first-token), `input_tokens`, `output_tokens`.

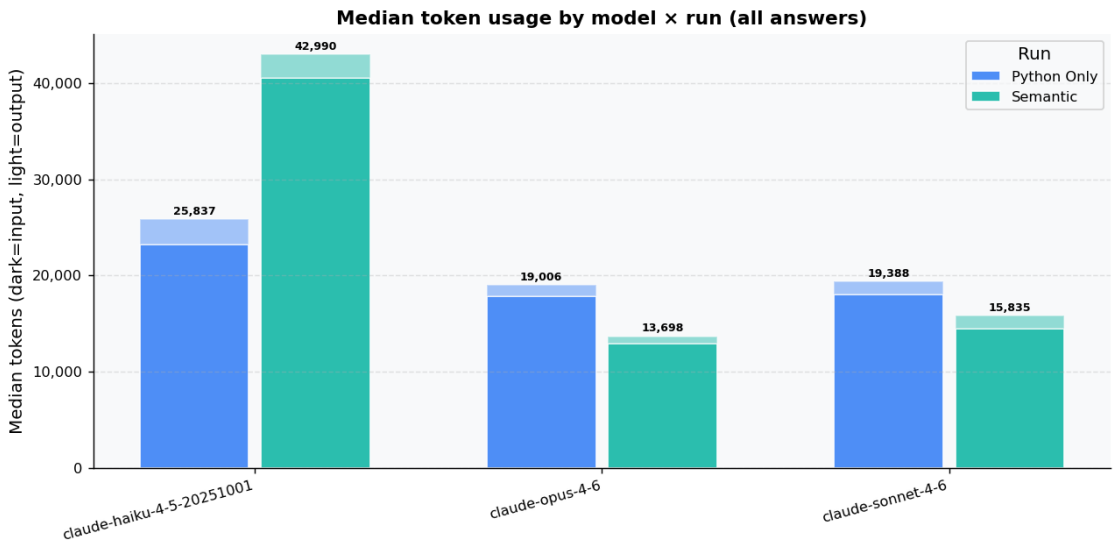


`No TTFT data available.`




```
Median metrics per run × model (errors excluded):

                                        n  total_latency_s ttft_s  input_tokens
output_tokens  total_tokens
run         model
Python Only claude-haiku-4-5-20251001  20            20.29   None      21243.50
2488.00      23664.00
            claude-opus-4-6            21            26.13   None      12543.00
920.00      13463.00
            claude-sonnet-4-6          19            24.26   None      16805.00
1229.00      18214.00
Semantic    claude-haiku-4-5-20251001  20            20.83   None      36628.50
2322.00      38647.50
            claude-opus-4-6            23            21.23   None      12646.00
797.00      13453.00
            claude-sonnet-4-6          23            29.55   None      14308.00
1356.00      15785.00
```

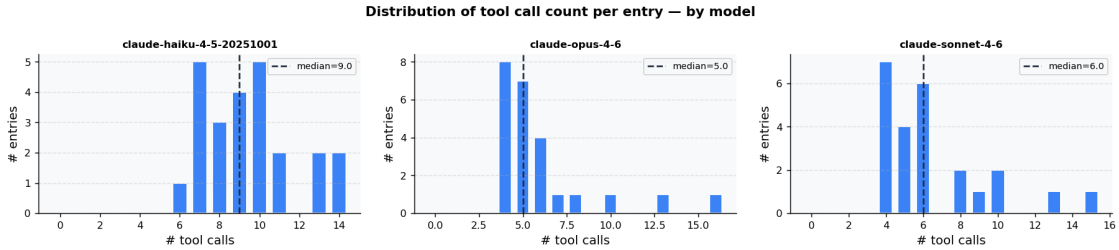


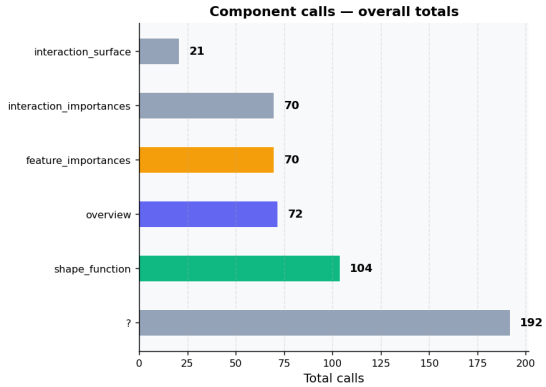


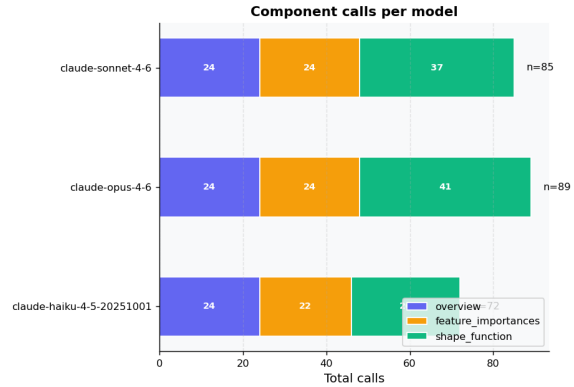
1.10 10 — Semantic Tool Usage Deep Dive

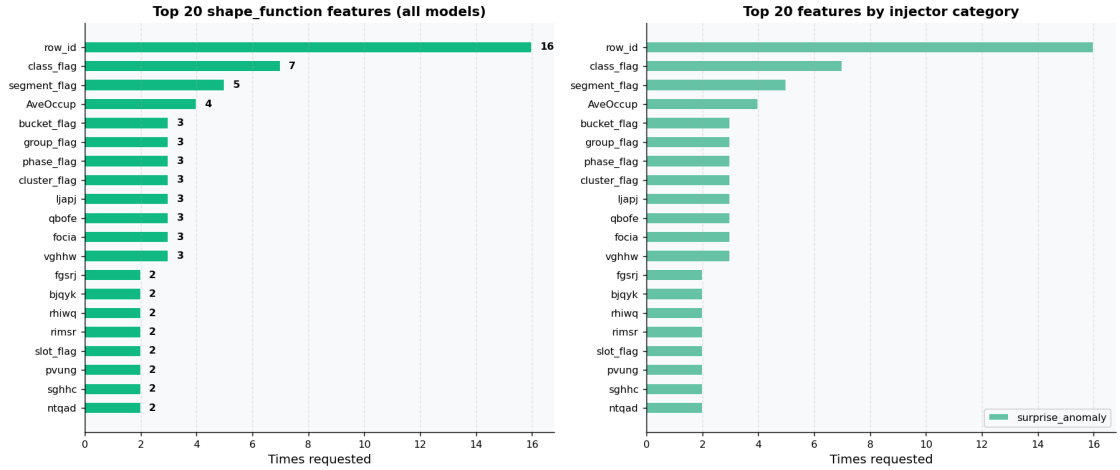
```
Entries in semantic run       : 72
Entries with 0 tool calls     : 0  (0.0%)
Total tool calls made         : 529
Median calls per entry        : 6.0
Max calls per entry           : 16

Calls per entry by model:
```

```
                          median  mean  max  zero_calls
model
claude-haiku-4-5-20251001    9.00  9.46   14           0
claude-opus-4-6              5.00  6.04   16           0
claude-sonnet-4-6            6.00  6.54   15           0

Component breakdown (all entries):
component
?                          192
shape_function             104
overview                    72
feature_importances         70
interaction_importances     70
interaction_surface         21
```

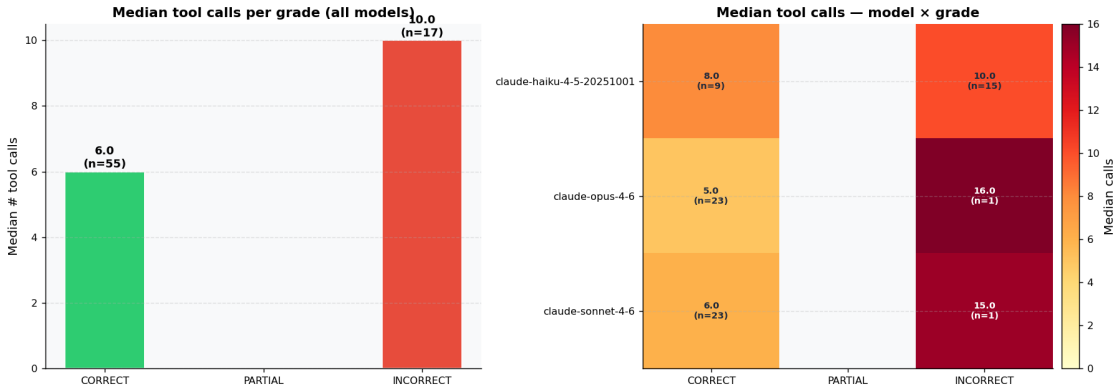









```
posx and posy should be finite values
posx and posy should be finite values
```

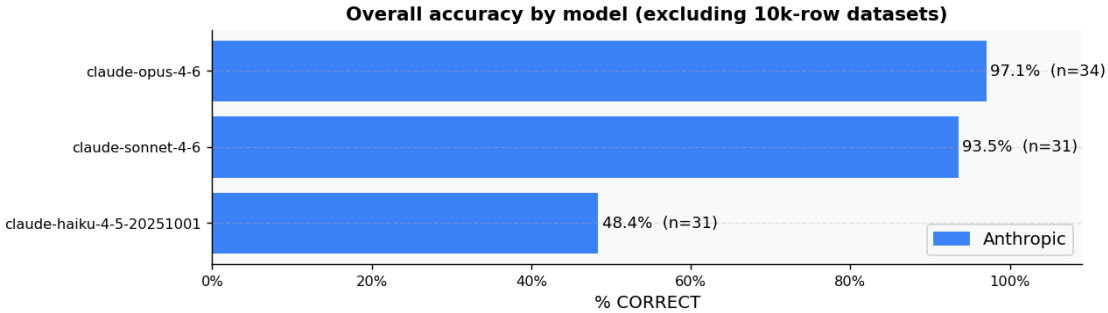


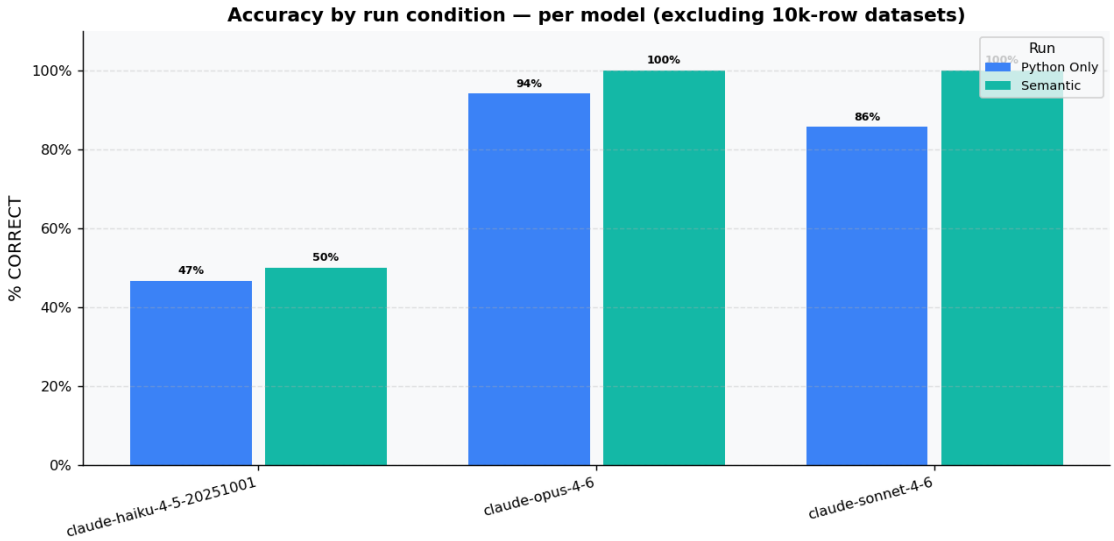

```
Entries where no tools were called: 0

No zero-call entries found.
```

1.11 11 — Performance Excluding 10k-Row Datasets

Since 10k-row datasets cause empty answers for csv_context (29% error rate), this section filters them out entirely so we can compare modes on an equal footing.

```
Records after excluding 10k-row datasets: 96 / 126
Datasets kept: ['bike_sharing_100', 'bike_sharing_1000', 'bike_sharing_500',
'california_housing_100', 'california_housing_1000', 'california_housing_500',
'vgsales_100', 'vgsales_1000', 'vgsales_500']
```


Overall accuracy by model (excluding 10k-row datasets)
claude-opus-4-6
claude-sonnet-4-6
claude-haiku-4-5-20251001
97.1% (n=34)
93.5% (n=31)
48.4% (n=31)
Anthropic
0%
20%
40%
60%
80%
100%
% CORRECT


Accuracy by run condition — per model (excluding 10k-row datasets)
Run
Python Only
Semantic
% CORRECT
100%
80%
60%
40%
20%
0%
47%
50%
94%
100%
86%
claude-haiku-4-5-20251001
claude-opus-4-6
claude-sonnet-4-6